

ROSHAN KUMAR GUPTA

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2. CAREER OBJECTIVES

Seeking a challenging position as a **Systems Engineer or Research Scientist** at a forward-thinking technology company to apply my expertise in high-performance computing, low-level systems, and technical leadership. I am a highly motivated professional committed to bridge the gap between complex hardware-software architectures and scalable, real-world solutions through effective team coordination and technical advocacy.

3. EXPERIENCES / EMPLOYMENT HISTORY

L3 IT Intern | SMG Electric Scooters Ltd, Chandigarh
Strategic Systems Optimization & CRM Development

Dec 2025 – Jan 2026

- Designed and developed a custom internal CRM portal, streamlining logistics and vendor development workflows for enterprise-scale operations.
- Collaborated with cross-functional teams to identify infrastructure bottlenecks, reducing manual processing time through automation.

Independent Research Engineer | High-Performance Computing
Technical Architecture & Multi-Disciplinary Research

2024 – Present

- Project TARS:** Successfully articulated research value to secure a \$5,500 Emergent Ventures grant; architected a physics-constrained ML pipeline achieving 70.1% precision.
- FieldChain:** Engineered a GPU-centric storage integrity layer; optimized PCIe 4.0 throughput to outperform specialized DPUs by 4.2x.
- HSA* Algorithm:** Formulated and communicated a novel routing engine achieving 140x faster pathfinding than standard Dijkstra implementations.
- Cryptic (ZeroToOne Winner): Frontend Lead** for a DeFi Treasury dashboard; managed UI/UX architecture and team integration for a high-priority hackathon winning project.

4. EDUCATIONAL QUALIFICATION

Bachelor of Technology in Computer Science | Chandigarh University, Mohali
Balancing a rigorous Computer Science curriculum with independent systems research and high-performance computing projects.

Expected 2029

5. AWARDS

- Emergent Ventures Grantee** (\$5,500) for high-impact research in Physics-ML.
- Winner**, ZeroToOne Hackathon (DeFi / Smart Contract Architecture).
- Winner**, Zinnovatio Hackathon (Rapid Prototyping & Deployment).
- Awarded multiple merit citations for high-performance systems engineering.

6. SKILL SETS AND PERSONAL ATTRIBUTES

Technical: CUDA, Vulkan, PTX, C++, Deterministic Runtimes, React Architecture, PyTorch.

Leadership: Technical Advocacy, Strategic Liaison, Cross-Functional Team Leadership, Scientific Writing, Stakeholder Management.

7. VOLUNTARY WORK / LEADERSHIP

Student Representative | Chandigarh University
Advocacy & Strategic Liaison

Oct 2025 – Sep 2026

- Acted as the primary liaison between the student body and university administration, managing high-stakes communication to resolve complex infrastructure and academic issues.
- Facilitated multiple departmental town-halls, ensuring transparent information flow and collaborative problem-solving across 400+ student cohorts.

8. EXTRACURRICULAR ACTIVITIES, INTERESTS AND HOBBIES

Highly active in the global hackathon circuit (15+ events), focus on building collaborative open-source tools, and exploring the intersection of quantum noise limits and signal processing.

9. REFERENCES

Available upon request from academic mentors and project leads.

10. SIGNATURE

Roshan Kumar Gupta